

TOKYO, Japan

WMO: 476620

Lat: 35.6917N

Lon: 139.7511E

Elev: 24

StdP: 101.03

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.7	1.5	-14.9	1.0	5.4	-13.4	1.2	5.3	8.9	6.8	8.1	7.1	3.1	340	0.409

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.1	33.7	25.6	32.6	25.2	31.5	24.7	26.6	31.3	26.0	30.5	25.5	29.8	4.0	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.4	20.7	29.0	24.8	19.9	28.7	24.2	19.2	28.5	83.5	31.3	80.9	30.6	78.7	29.9	29.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.2	7.1	6.3	-1.0	36.1	1.0	1.4	-1.6	37.2	-2.2	38.0	-2.7	38.8	-3.4	39.8		
			-4.3	27.4	0.8	0.8	-4.9	28.0	-5.4	28.5	-5.8	29.0	-6.4	29.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.8	6.2	6.8	9.9	15.0	19.6	22.6	26.8	28.0	24.4	19.1	13.6	8.6
	DBStd	7.97	2.05	2.57	3.18	3.28	2.67	2.61	2.91	2.53	3.16	2.80	2.72	2.44
	HDD10.0	322	120	96	43	3	0	0	0	0	0	3	57	
	HDD18.3	1559	377	324	262	107	18	2	0	1	23	144	301	
	CDD10.0	2791	2	6	39	153	297	378	521	558	432	282	110	14
	CDD18.3	986	0	0	0	7	56	130	262	300	183	47	1	0
	CDH23.3	8478	0	0	1	17	192	672	2656	3379	1412	149	1	0
	CDH26.7	2953	0	0	0	1	22	132	993	1392	397	17	0	0
Wind	WSAvg	3.1	3.1	3.3	3.4	3.5	3.3	2.9	3.0	3.0	3.1	3.0	2.8	2.9
Precipitation	PrecAvg	1548	62	64	112	124	137	160	149	145	218	209	108	61
	PrecMax	2040	141	141	198	234	256	319	325	356	462	683	256	198
	PrecMin	1142	3	4	36	44	62	44	31	14	46	18	8	1
	PrecStd	219	40	36	39	51	50	58	92	95	104	147	60	41
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.6	18.5	21.3	25.3	29.1	31.5	34.8	35.5	33.1	28.7	22.5	18.7
		MCWB	10.5	12.4	13.3	16.4	19.1	23.1	25.4	25.7	24.3	21.3	15.9	13.3
	2%	DB	13.2	15.2	19.1	23.2	26.7	29.6	33.5	34.0	31.5	26.4	20.9	15.9
		MCWB	7.6	9.5	12.5	15.6	18.6	22.0	25.4	25.8	24.1	20.2	15.1	9.8
	5%	DB	11.5	12.9	17.3	21.7	25.4	28.1	32.4	33.0	30.2	25.0	19.4	14.3
		MCWB	5.7	7.6	11.6	15.2	17.9	21.5	25.1	25.6	23.8	19.4	14.1	8.9
	10%	DB	10.2	11.3	15.6	20.4	24.1	26.8	31.2	31.9	28.9	23.5	18.1	12.9
		MCWB	4.6	5.8	10.3	14.1	17.4	21.0	24.7	25.2	23.2	18.2	13.1	7.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.7	14.2	15.8	20.1	22.1	24.7	27.3	27.8	26.2	23.2	18.2	14.5
		MCDB	14.8	17.9	18.5	21.7	25.5	29.3	32.3	32.9	30.2	26.3	20.8	17.4
	2%	WB	8.4	10.4	14.2	17.7	20.6	23.7	26.3	26.9	25.3	21.9	16.5	11.6
		MCDB	12.0	14.2	17.8	21.5	24.1	27.9	31.1	31.6	29.1	25.2	19.2	14.8
	5%	WB	6.7	8.2	12.7	16.3	19.7	22.9	25.8	26.3	24.8	20.6	15.4	9.8
		MCDB	10.5	11.8	16.5	20.3	23.4	26.6	30.4	30.8	28.5	23.7	18.1	13.1
	10%	WB	5.4	6.7	10.8	15.3	18.7	22.1	25.3	25.8	24.3	19.2	14.2	8.5
		MCDB	9.3	10.4	14.7	19.0	22.5	25.5	29.7	30.2	28.0	22.4	17.1	12.0
Mean Daily Temperature Range	5% DB	MDBR	6.8	7.0	7.4	7.5	7.0	5.9	5.9	6.1	5.7	5.8	6.4	6.6
		MCDBR	8.9	9.5	9.5	9.1	8.8	7.6	7.3	7.2	6.8	7.4	8.1	8.6
		MCWBR	6.1	7.0	6.6	5.2	4.4	3.2	2.8	2.6	2.9	4.4	5.6	6.0
	5% WB	MCDBR	8.3	8.9	8.8	7.9	7.5	6.3	6.8	6.8	6.2	6.7	7.1	7.8
		MCWBR	6.4	7.3	7.2	5.6	4.7	3.2	2.8	2.8	3.0	4.6	5.8	6.3
Clear-Sky Solar Irradiance	taub	0.380	0.429	0.497	0.548	0.572	0.608	0.600	0.588	0.520	0.488	0.428	0.383	
	taud	2.238	2.096	1.937	1.858	1.857	1.837	1.905	1.922	2.074	2.090	2.171	2.252	
	Ebn at Noon	794	792	769	753	741	713	715	713	742	724	731	762	
	Edh at Noon	107	139	177	202	204	208	194	187	153	138	115	100	
All-Sky Solar Radiation	RadAvg	2.77	3.28	4.09	4.77	5.13	4.52	4.90	4.93	3.83	3.03	2.51	2.42	
	RadStd	0.19	0.32	0.26	0.47	0.54	0.42	0.82	0.47	0.34	0.32	0.24	0.18	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page